

# James Hutchins

30 Colborne St. Elora Ontario, N0B1S0

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## Professional Summary

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I am a full-stack software developer and data engineer with dual degrees in Computer Science and Mechanical Engineering. I specialize in building artificial intelligence agent systems, engineering robust data pipelines, and deploying scalable web applications. Drawing on my experience as a startup founder and enterprise data analyst, I focus on translating complex technical requirements into reliable production software. My primary objective is always to solve tangible business problems, automate operational workflows, and drive measurable financial value.

## Education

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| <b>Western University</b> , London, ON, Canada<br><i>BESc, Mechanical, and Materials Engineering</i> | Graduated 2024 |
| <b>Western University</b> , London, ON, Canada<br><i>BSc Computer Science</i>                        | Graduated 2025 |

## Relevant Coursework

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| <ul style="list-style-type: none"><li>Macro &amp; Microeconomics</li><li>Machine Learning</li></ul> | <ul style="list-style-type: none"><li>Software Engineering</li><li>Databases</li></ul> | <ul style="list-style-type: none"><li>Software Project Management</li></ul> |
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## Experience

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| <b>Bank Of Montreal</b><br><i>Data Analyst</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Toronto, Ontario, Canada — Winter 2025       |
| <ul style="list-style-type: none"><li>Partnered with internal stakeholders to drive data quality analytics and issue triage operations.</li><li>Built Python and Azure ETL pipelines with strict validation rules to improve downstream data reliability.</li><li>Developed Power BI dashboards and SQL reporting to surface operational trends for business partners.</li><li>Automated enterprise workflows with cloud scripts, directly reducing manual labor and accelerating delivery.</li><li>Optimized data validation pipelines processing 500 million rows daily, focusing on system robustness.</li></ul>                                                                                                                                                                                                                                       |                                              |
| <b>Dialogix Services Inc</b><br><i>Co-Founder</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Toronto, Ontario, Canada — June 2025-Present |
| <ul style="list-style-type: none"><li>Built standardized CI/CD pipelines on Google Cloud to accelerate deployments and ensure system reliability.</li><li>Managed deployment and infrastructure through Google Cloud Platform to maintain high application uptime.</li><li>Implemented automated testing and quality gates to minimize regressions and increase release confidence.</li><li>Engineered structured logging and error detection workflows to accelerate issue discovery during support.</li><li>Co-authored and shipped a production web application currently used by the Ivey Business School.</li><li>Maintained live production codebases by rapidly triaging issues, executing QA, and delivering bug fixes.</li><li>Collaborated with stakeholders to translate complex business requirements into shipped production code.</li></ul> |                                              |
| <b>Blue Guardian</b> , London, Ontario, Canada — Winter 2023<br><i>Machine Learning NLP R&amp;D Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                              |
| <ul style="list-style-type: none"><li>Built a BERT natural language pipeline to detect risk signals for a live mobile software application.</li><li>Enhanced text classification by upgrading traditional statistical models to a transformer architecture.</li><li>Curated training datasets and documentation to ensure reliable model retraining and quality control.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                              |
| <b>Vector Institute</b><br><i>Generative AI (Artificial Intelligence) Research Intern</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | London, Ontario, Canada — Summer 2024        |
| <ul style="list-style-type: none"><li>Engineered PyTorch machine learning pipelines for dataset preparation, feature engineering, and training.</li><li>Leveraged transformer and diffusion models to decode participant EEG signals into generated images.</li><li>Managed large-scale model training runs on compute clusters with strict monitoring and quality controls.</li><li>Collaborated with researchers to iterate quickly on hypotheses, baselines, and ablation comparisons.</li></ul>                                                                                                                                                                                                                                                                                                                                                       |                                              |

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## Scotiabank

Toronto, Ontario, Canada — Summer 2023

### *Small Business Analyst*

- Performed financial risk assessments by analyzing cash flow, industry trends, and client documentation.
- Improved data consistency for credit pricing reporting by implementing strict validation processes.
- Built VBA automation to standardize customer assessments, reducing manual effort across the analytics team.
- Produced pricing recommendations and competitive research briefs to support underwriting decisions.

## Projects

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### Zenith Vision Analytics,

Ontario, Canada — 2024-Present

#### Founder

- Built an AI agent orchestrator using LangChain and MCP to automate customer support workflows.
- Developed voice copilots that reduced manual onboarding and support labor for business owners.
- Shipped a functional AI platform to paying customers, driving revenue and lowering operational costs.
- Built AWS Apache Airflow pipelines for automated data collection, quality validation, and refresh scheduling.
- Engineered PostgreSQL databases to support complex data ingestion, enrichment, and analytics workloads.
- Processed TB-scale imagery datasets using YOLO and SAM computer vision models to generate structured data.
- Created geospatial datasets by cleaning data from multiple sources to enable property insights.
- Implemented PostgreSQL vector search to power product comparison workflows and improve matching relevance.
- Delivered BI dashboards that translated collected data into actionable recommendations and ROI clarity.

## Technical Skills

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**Languages:** Python, JavaScript, SQL, C/C++, Java, R

**Machine Learning & AI:** PyTorch, NLP, Computer Vision (YOLO, SAM), Large Language Models, Vector Search, Langchain

**Backend & Data:** REST APIs (FastAPI, Flask), PostgreSQL, MongoDB, ETL Pipelines, Pandas, Scikit-learn

**Infrastructure:** Linux, Docker, Google Cloud Platform, AWS, Azure, Git, Apache Airflow, CI/CD